

MARCO ANTONIO ADAME RODRÍGUEZ

IoT & Embedded Systems Engineer · Digital Signal Processing · Scientific Software

Mexico (GMT-6) · Available for remote work with US and European overlap

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PROFILE

Computer Engineer and MSc candidate in Engineering, Innovation and Technological Development (Universidad Autónoma de Guerrero, expected November 2026). I build end-to-end IoT systems: MEMS sensor, embedded firmware, real-time acquisition, Linux backend, cloud storage and visualization dashboard. My graduate work is a low-cost structural health monitoring (SHM) system I designed from scratch and validated in the field against commercial seismic instrumentation. Lead author of a peer-reviewed paper in Revista Vínculos (Universidad Distrital, Colombia), co-author of two more, and speaker at the 15th International Computing Conference (Tunja, Colombia, 2025). Daily work in scientific Python (NumPy, SciPy, Pandas), with services deployed on Ubuntu Server, Docker and the cloud.

KEY ACHIEVEMENTS

- **Complete SHM-IoT system designed from the ground up** and validated in the field against commercial GEA and P-Alert equipment, at a fraction of the cost of traditional seismic instrumentation.
- **Ambient vibration measurement system on a biaxial shaking table:** synchronized 200 Hz sampling, FFT analysis over 2048-point windows, 0.98 Hz frequency resolution, and automatic RPM computation from the dominant frequency component (peer-reviewed publication).
- **Continuously running multi-site monitoring infrastructure:** nodes across separate networks, a centralized Ubuntu Server receiver, local buffering against connectivity loss, and published web dashboards.
- **Python/PyQt5 vibration analysis suite** that processes large volumes of records and generates comparative technical reports automatically.

RESEARCH & DEVELOPMENT EXPERIENCE

Lead Researcher and Developer — SHM-IoT System · UAGro · MIIDT program

Aug 2024 – Present

End-to-end design and implementation of a low-cost structural health monitoring solution for civil works, as an accessible alternative to commercial seismic instrumentation.

Acquisition, firmware and hardware

- Integrated acquisition nodes on Arduino Nano RP2040 Connect and Raspberry Pi 4 with MPU9250/MPU6050, LSM6DSOX and ADXL335 MEMS sensors over I2C, SPI and UART.
- Implemented synchronized acquisition on a dual-core architecture, sampling at up to 200 Hz and satisfying the Nyquist criterion for the structural frequency range of interest.
- Validated the system on a biaxial shaking table, university buildings, a botanical garden, a high school campus and lightweight experimental structures, benchmarking it directly against commercial GEA and P-Alert units.

Digital signal processing and scientific software

- Built the system's DSP pipeline: RMS, FFT, power spectral density (Welch's method), dominant frequency, coherence, phase analysis and cross-correlation between sensors.
- Developed Python and PyQt5 desktop applications for acquisition, live monitoring and analysis, with session control, result export and automatic crash recovery.
- Created tooling to process TXT, CSV, JSON and JSONL files with large record volumes, including time synchronization, resampling, gap detection, digital filtering and series trimming.
- Automated the generation of technical reports and comparative plots documenting system performance against reference equipment.

Backend, cloud and infrastructure

- Built a centralized receiver that integrates sensors located at different sites and networks over HTTP/HTTPS, with multi-source storage and local buffering that prevents data loss during connectivity outages.
- Developed REST APIs in FastAPI for ingesting, persisting and querying telemetry sent by ESP32 microcontrollers.
- Administered continuously running Ubuntu servers: systemd auto-start services, remote access over SSH/VNC, Docker containerization, and backup and synchronization strategies.
- Published web dashboards and query services that allow users to visualize, filter and download data from both custom nodes and commercial sensors.

PUBLICATIONS & CONFERENCE TALKS

- Adame Rodríguez, M. A., Alcaraz Vázquez, M., Martínez, E. T., García Guevara, D. S., & Alonso Silverio, G. A. (2026). Ambient Vibration Measurement System on a Biaxial Vibrating Table. Revista Vínculos, 22(1). [Lead author]
- Alcaraz Vázquez, M., Adame Rodríguez, M. A., García Guevara, D. S., Martínez, E. T., & Alonso Silverio, G. A. (2025). Feature-importance-based attribute selection for dengue case classification in Mexico. Revista Vínculos, 22(2). [Co-author]

- García Guevara, D. S., Martínez, E. T., Alcaraz Vázquez, M., Adame Rodríguez, M. A., & Alonso Silverio, G. A. (2025). IoT-based biological sample shaker with temperature control. *Revista Vínculos*, 22(2). [Co-author]
- **Conference talk:** "Ambient Vibration Measurement System on a Biaxial Vibrating Table". 15th International Computing Conference (CICOM 2025), Tunja, Colombia, October 2025.

TECHNICAL SKILLS

Languages: Python, C/C++ (embedded), JavaScript, SQL, PHP, Bash.

Data & signals: NumPy, Pandas, SciPy, Matplotlib · FFT, PSD/Welch, RMS, digital filters, coherence, phase analysis, resampling, time series.

IoT & hardware: Raspberry Pi, ESP32, Arduino Nano RP2040 Connect · MEMS sensors (IMU/accelerometers) · I2C, UART, SPI, WiFi, Ethernet · real-time acquisition.

Backend & cloud: FastAPI, REST APIs, HTTP/HTTPS, webhooks · Supabase, PostgreSQL, Firebase · distributed sync and storage.

Systems & DevOps: Linux (Ubuntu Server/Desktop), Docker, systemd, SSH, VNC, Git/GitHub, Vercel, service automation.

Interfaces & visualization: PyQt5, web dashboards, scientific visualization, CSV/JSON/PDF export.

SELECTED PROJECTS

SHM-IoT MIIDT — Low-cost structural monitoring · Python, C++, Raspberry Pi, PyQt5

Full architecture for acquisition, storage, analysis and visualization of accelerations in civil structures and experimental models, field-tested against commercial sensors.

Geo-referenced SHM monitoring network · JavaScript, Vercel

Web dashboard showing the location and status of distributed monitoring nodes, continuously deployed and publicly available.

github.com/Ryukert/SHM_GPS_NETWORK_DASH_elite

Supabase sensor dashboard · JavaScript, Supabase, PostgreSQL

Web application for submitting, querying and visualizing measurements stored in the cloud. github.com/Ryukert/MIIDT_SUPABASE_DASH

ESP32 telemetry APIs · Python, FastAPI

REST services for receiving, persisting and querying acceleration data, plus scientific plotting tools. github.com/Ryukert/APIAceleraciones

EDUCATION

MSc in Engineering, Innovation and Technological Development · UAGro

Aug 2024 – Nov 2026 (*exp.*)

Thesis: "Development and implementation of a low-cost IoT system for structural health monitoring in civil works". In final stage.

BSc in Computer Engineering · UAGro

2018 – 2022 · Degree awarded 2023

Federal professional license (cédula profesional) no. 13426767.

ADDITIONAL EXPERIENCE

- **Professional internship — IT area, 35th Military Zone (Mexican Army)** · Aug 2022 – Jan 2023. 450 certified hours. Computer equipment support and maintenance in an institutional environment with strict operational discipline and sensitive-information handling.
- **Social service — Ignacio Manuel Altamirano Secondary School** · Feb 2022 – Jul 2022. 500 certified hours. Technical support in the computing area and community-sector educational support.

CERTIFICATIONS & LANGUAGES

- Data Science Essentials with Python — Cisco Networking Academy / UAGro (Dec 2025).
- Cybersecurity Essentials — Cisco Networking Academy (Mar 2022).
- NDG Linux Unhatched — Cisco Networking Academy (May 2022).
- CCNAv7: Introduction to Networks — Cisco Networking Academy (Jun 2021).
- **Languages:** Spanish (native). English intermediate (B1), TOEFL ITP 470 (2025); fluent reading of technical documentation and scientific literature.